

- [8] X. D. Tang, G. Tao, and S. M. Joshi, "Adaptive actuator failure compensation for parametric strict feedback systems and an aircraft application," *Automatica*, vol. 39, no. 11, pp. 1975–1982, 2003.
- [9] X. D. Tang, G. Tao, and S. M. Joshi, "Adaptive output feedback actuator failure compensation for a class of nonlinear systems," *Int. J. Adapt. Control and Signal Process.*, vol. 19, pp. 419–444, 2005.
- [10] W. Wang and C. Wen, "A new approach to adaptive actuator failure compensation in uncertain systems," in *Proc. IEEE Int. Conf. Control and Automation (ICCA 2009)*, 2009, pp. 47–52.
- [11] W. Wang and C. Wen, "Adaptive output feedback controller design for a class of uncertain nonlinear systems with actuator failures," in *Proc. 49th IEEE Conf. Decision and Control 2010*, pp. 1749–1754.
- [12] W. Wang and C. Wen, "Adaptive actuator failure compensation control of uncertain nonlinear systems with guaranteed transient performance," *Automatica*, vol. 46, no. 12, pp. 2082–2091, 2010.
- [13] J. P. Cai, C. Wen, H. Y. Su, X. D. Li, and Z. T. Liu, "Adaptive failure compensation of hysteretic actuators in controlling uncertain nonlinear systems," in *Proc. 2011 ACC*, 2011, pp. 2320–2325.
- [14] J. P. Cai, C. Wen, H. Y. Su, and Z. T. Liu, "Robust adaptive failure compensation of hysteretic actuators for parametric strict feedback systems," in *Proc. 2011 CDC-ECC*, 2011, pp. 7988–7993.
- [15] M. Krstic, I. Kanellakopoulos, and P. V. Kokotovic, *Nonlinear and Adaptive Control Design*. New York: Wiley, 1995.
- [16] J. Zhou and C. Wen, *Adaptive Backstepping Control of Uncertain Systems*. Berlin/Heidelberg: Springer-Verlag, 2007.

A Robust MPC Algorithm for Offset-Free Tracking of Constant Reference Signals

Giulio Betti, *Graduate Student Member, IEEE*, Marcello Farina, and Riccardo Scattolini

Abstract—A robust model predictive control algorithm solving the tracking and the infeasible reference problems for constrained systems subject to bounded disturbances is presented in this technical note. The proposed solution relies on three main concepts: 1) the reformulation of the system in the so-called velocity form to obtain offset-free tracking when constant disturbances are present, 2) the use of a tube-based approach to cope with non-constant but bounded disturbances, 3) the use of reference outputs as arguments of the optimization problem to cope with infeasible references. Convergence results are derived by suitably defining the auxiliary control law and the terminal set used in the problem formulation.

Index Terms—Constrained system, disturbance rejection, model predictive control (MPC), robust MPC, tracking.

I. INTRODUCTION

This technical note proposes a model predictive control (MPC) method for tracking constant reference signals. The controlled system is assumed to be described by a discrete-time linear model affected by unknown bounded disturbances. The proposed solution relies on the robust "tube-based" control design approach, first discussed in [14], to guarantee convergence properties in the face of unknown exogenous

signals, and on the reformulation of the system model in the so-called *velocity form*, see [2], [20], [21], [24]. The problem of infeasible references is solved following the approach recently proposed in [1], [9], [10].

Different solutions have been proposed for the design of MPC algorithms for tracking. Many methods, see, e.g., [11], [15], [16], [17], [19], require to augment the model with an artificial disturbance to be estimated together with the system state. Another approach directly forces an integrator in the control scheme; this strategy has been followed in [12], [13]. A third possible solution consists of reformulating the system in the so-called "velocity form," see [2], [20], [21], [24], where the enlarged state is composed by the state increments and the output error, while the manipulated variable is the control increment. The use of the velocity form can be advantageous, since it does not require to compute the steady state target in order to properly formulate the optimization problem considered in the MPC formulation. In order to address the stability issue with MPC algorithms for systems in velocity form, in [24] the "zero terminal constraint" solution is suggested. However, this solution cannot be used for systems affected by bounded but non-constant disturbances and, to the best of the authors' knowledge, no general stability results have been presented so far for this method, especially for the robust case.

The problems of infeasible references has been discussed in, e.g., [3], [18], [23] and an efficient solution has been recently proposed in [1], [9], [10], where the MPC cost function is complemented by a term explicitly penalizing the distance between the required (possibly infeasible) reference signal **and an artificial, but always feasible, reference, which turns out to be one of the optimization variables.**

In this technical note we propose a novel control scheme that resorts to "tube-based" MPC for coping with disturbances and uncertainties, based on the velocity form for tracking constant reference signals, and able to cope with infeasible reference problems. The proposed control scheme is used in three cases, which differ from each other in view of the type of perturbation affecting the system, i.e., bounded disturbance, constant disturbance, and model uncertainty.

The technical note is organized as follows. In Section II the real and nominal plant models are introduced, and the corresponding velocity forms are defined. Moreover, the constraints on the original state and control variables are transformed into equivalent constraints on the variables of the velocity form and some additional design parameters are specified. Section III is devoted to presenting the robust MPC algorithm together with its convergence properties and the three cases mentioned above. In Section IV a simulation example is reported while some concluding remarks are drawn in Section V. All the proofs are reported in the Appendix to improve readability.

Notation: A matrix is Schur if all its eigenvalues lie in the interior of the unit circle. The short-hand $v = (v_1, \dots, v_s)$ denotes the column vector $v = [v_1^T, \dots, v_s^T]^T$ with s (possibly vector) components. For a discrete-time signal s_t and $a, b \in \mathbb{N}$, $a \leq b$, we denote $(s_a, s_{a+1}, \dots, s_b)$ with $s_{[a:b]}$. The symbol $\oplus$ denotes the Minkowski sum, while $\ominus$ stands for the Pontryagin difference see, e.g., [14]. $B_\varepsilon^{dim}(0)$ defines a ball of radius ε in the $\mathbb{R}^{dim}$ space. Finally, I_n is the $n \times n$ identity matrix, and 0 is the matrix with zero entries, whose dimensions are clear from the context.

II. STATEMENT OF THE PROBLEM

Consider the discrete-time, linear, time-invariant system, defined as

$$\begin{aligned} x_{t+1} &= Ax_t + Bu_t + w_t \\ y_t &= Cx_t \end{aligned} \quad (1)$$

Manuscript received June 21, 2012; revised October 23, 2012, February 21, 2013; accepted March 06, 2013. Date of publication March 22, 2013; date of current version August 15, 2013. Recommended by Associate Editor D. Arzeller.

The authors are with the Dipartimento di Elettronica, Informazione e Bioingegneria, Politecnico di Milano, Milan, Italy (e-mail: betti@elet.polimi.it; farina@elet.polimi.it; riccardo.scattolini@elet.polimi.it).

Digital Object Identifier 10.1109/TAC.2013.2254011

where $x_t \in \mathbb{X} \subseteq \mathbb{R}^n$, $u_t \in \mathbb{U} \subseteq \mathbb{R}^m$, $y_t \in \mathbb{R}^p$ are the state, input, and output vectors, while $\mathbb{X}, \mathbb{U}$ are compact and convex sets. The disturbance $w_t \in \mathbb{W}$ is unknown, but bounded, and $\mathbb{W}$ includes the origin. Concerning system (1), the following assumptions are made.

Assumption 1:

- (i) The state is measurable.
- (ii) $m = p$.
- (iii) The pair (A, B) is reachable.
- (iv) $\text{rank}(S) = n + p$, where

$$S = \begin{bmatrix} I_n - A & -B \\ C & 0 \end{bmatrix}.$$

Condition 1 (ii) could be relaxed to $m \geq p$. However, square systems are considered in the following to simplify the mathematical developments.

A. The Velocity Form and the Auxiliary Control Law

In order to design an MPC state-feedback control law for tracking a given constant reference signal $r^o \in \mathbb{R}^m$, and according to [2], [20], [21], [24], system (1) is enlarged with m integrators and described in the so-called *velocity form*. Specifically, let $\hat{r}$ be a generic tracking target (which could be different from r^o for reasons which will become clear later on). Then, letting $\delta x_t = x_t - x_{t-1}$, $\varepsilon_t = y_t - \hat{r}$, $\delta u_t = u_t - u_{t-1}$, and $\delta w_t = w_t - w_{t-1}$, system (1) can be reformulated as follows:

$$\delta x_{t+1} = A\delta x_t + B\delta u_t + \delta w_t \quad (2)$$

$$\varepsilon_{t+1} = CA\delta x_t + \varepsilon_t + CB\delta u_t + C\delta w_t.$$

Define $\xi_t = (\delta x_t, \varepsilon_t)$, and

$$\mathcal{A} = \begin{bmatrix} A & 0 \\ CA & I_m \end{bmatrix}, \mathcal{B} = \begin{bmatrix} B \\ CB \end{bmatrix}, \mathcal{B}_w = \begin{bmatrix} I_n \\ C \end{bmatrix}, d_t = \mathcal{B}_w \delta w_t \quad (3) \text{ and}$$

and write system (2) in compact form as

$$\xi_{t+1} = \mathcal{A}\xi_t + \mathcal{B}\delta u_t + d_t. \quad (4)$$

The following proposition can be easily proved.

Proposition 1: If Assumption 1 holds, then the pair $(\mathcal{A}, \mathcal{B})$ is reachable. ■

Associated with system (1), the following *nominal* system is introduced:

$$\begin{aligned} \hat{x}_{t+1} &= A\hat{x}_t + B\hat{u}_t \\ \hat{y}_t &= C\hat{x}_t. \end{aligned} \quad (5)$$

Letting $\delta \hat{x}_t = \hat{x}_t - \hat{x}_{t-1}$, $\hat{\varepsilon}_t = \hat{y}_t - \hat{r}$, $\delta \hat{u}_t = \hat{u}_t - \hat{u}_{t-1}$, $\hat{\xi}_t = (\delta \hat{x}_t, \hat{\varepsilon}_t)$, the nominal counterparts of systems (2), (4) are

$$\begin{aligned} \delta \hat{x}_{t+1} &= A\delta \hat{x}_t + B\delta \hat{u}_t \\ \hat{\varepsilon}_{t+1} &= CA\delta \hat{x}_t + \hat{\varepsilon}_t + CB\delta \hat{u}_t \end{aligned} \quad (6)$$

and

$$\hat{\xi}_{t+1} = \mathcal{A}\hat{\xi}_t + \mathcal{B}\delta \hat{u}_t. \quad (7)$$

Assume that, for all t , for the real system (4) the following control law is considered:

$$\delta u_t = \delta \hat{u}_t + \mathcal{K}(\xi_t - \hat{\xi}_t) \quad (8)$$

where the gain $\mathcal{K}$ is defined *a priori* such that $\mathcal{F} = \mathcal{A} + \mathcal{B}\mathcal{K}$ is Schur (see Proposition 1). From (4), (7), and (8) it directly follows that:

$$(\xi_{t+1} - \hat{\xi}_{t+1}) = \mathcal{F}(\xi_t - \hat{\xi}_t) + d_t \quad (9)$$

which is independent of the control input $\delta \hat{u}_t$ to the nominal system.

B. Tightened Constraints

One of the main obstacles for the use of the velocity form is due to the difficulty to transform the state and control constraints in terms of equivalent constraints on the enlarged nominal state $\hat{\xi}_t$ and of the reference signal $\hat{r}$. To this end, the approach outlined in the following consists in three logical steps:

- (I) write $(x_t - \hat{x}_t, u_{t-1} - \hat{u}_{t-1})$ in terms of $(\xi_t - \hat{\xi}_t, w_{t-1})$;
- (II) show that there exists a robust positively invariant (RPI) set (denoted $\tilde{\mathbb{Z}}$) where $(\xi_t - \hat{\xi}_t, w_{t-1})$ is guaranteed to lie;
- (III) compute, from $\tilde{\mathbb{Z}}$, the tightened set $\tilde{\mathbb{X}}_{\mathbb{U}}$ where $(\hat{x}_t, \hat{u}_{t-1})$ must lie to verify the constraints on x_t and u_t .

I) The following result will be used.

Proposition 2: Under Assumption 1, the following equations hold:

$$\begin{bmatrix} x_t \\ u_{t-1} \end{bmatrix} = C^* \begin{bmatrix} \xi_t \\ \hat{r}_t \end{bmatrix} + C_w w_{t-1} \quad (10)$$

$$\begin{bmatrix} \hat{x}_t \\ \hat{u}_{t-1} \end{bmatrix} = C^* \begin{bmatrix} \hat{\xi}_t \\ \hat{r}_t \end{bmatrix} \quad (11)$$

where

$$\begin{aligned} C^* &= \begin{bmatrix} A & B \\ 0 & I_m \end{bmatrix} \Sigma^{-1} \begin{bmatrix} I_n & 0 & 0 \\ 0 & I_m & I_m \end{bmatrix} \\ C_w &= \begin{bmatrix} I_n \\ 0 \end{bmatrix} - \begin{bmatrix} A & B \\ 0 & I_m \end{bmatrix} \Sigma^{-1} \begin{bmatrix} I_n \\ C \end{bmatrix} \end{aligned}$$

$$\Sigma = \begin{bmatrix} A - I_n & B \\ CA & CB \end{bmatrix}$$

Note that the inverse of Σ always exists in view of Assumption 1 (ii) and (iv) (see Appendix A). Matrix C^* can be partitioned as follows: $C^* = [C_\xi \ C_y]$, where $C_\xi \in \mathbb{R}^{(n+m) \times (n+m)}$ and $C_y \in \mathbb{R}^{(n+m) \times m}$. Correspondingly, from (10), (11) the difference between real and nominal states and inputs is

$$\begin{bmatrix} x_t - \hat{x}_t \\ u_{t-1} - \hat{u}_{t-1} \end{bmatrix} = [C_\xi \ C_w] \begin{bmatrix} \xi_t - \hat{\xi}_t \\ w_{t-1} \end{bmatrix}. \quad (12)$$

II) Defining $\omega_t = w_{t-1}$ one can write, from (3) and (9)

$$\begin{bmatrix} \xi_{t+1} - \hat{\xi}_{t+1} \\ \omega_{t+1} \end{bmatrix} = \tilde{\mathcal{F}} \begin{bmatrix} \xi_t - \hat{\xi}_t \\ \omega_t \end{bmatrix} + \tilde{\mathcal{B}}_w w_t \quad (13)$$

where

$$\tilde{\mathcal{F}} = \begin{bmatrix} \mathcal{F} & -\mathcal{B}_w \\ 0 & 0 \end{bmatrix}, \quad \tilde{\mathcal{B}}_w = \begin{bmatrix} \mathcal{B}_w \\ I_n \end{bmatrix}.$$

From (13), since the Schureness of $\tilde{\mathcal{F}}$ follows from the Schureness of $\mathcal{F}$, in view of [22] it is possible to compute the minimal RPI set $\tilde{\mathbb{Z}}$ for $(\xi_t - \hat{\xi}_t, \omega_t)$ as:

$$\tilde{\mathbb{Z}} = \bigoplus_{k=0}^{\infty} \tilde{\mathcal{F}}^k \tilde{\mathcal{B}}_w \mathbb{W}. \quad (14)$$

Therefore, in view of (14), if $(\xi_t - \hat{\xi}_t, \omega_t) \in \tilde{Z}$ and $w_{t+j} \in W$ for all $j \geq 0$, it holds that $(\xi_{t+j}, \omega_{t+j}) \in (\hat{\xi}_{t+j}, 0) \oplus \tilde{Z}$ for all $j \geq 1$. An algorithm for the computation of a polytopic robustly invariant outer approximation of the latter set can be devised in line with [22].

Remark 1: an alternative choice is to set $d_t \in \mathbb{D} = \mathcal{B}_w(W \oplus (-W))$ in (4) and to compute a RPI set for (9) as $Z = \bigoplus_{k=0}^{\infty} \mathcal{F}^k \mathbb{D}$, such that if $(\xi_t - \hat{\xi}_t) \in Z$ then $\xi_{t+j} \in \hat{\xi}_{t+j} \oplus Z$ for all $j \geq 1$ and for all possible disturbance realizations. Such a simpler but more conservative choice corresponds to replace $\tilde{Z}$ in (14) with $Z \times W$ in the following developments.

III) Since $\omega_t = w_{t-1}$ and in view of (14) one has that, if $(\xi_t - \hat{\xi}_t, \omega_t) \in \tilde{Z}$, then

$$\begin{bmatrix} x_t \\ u_{t-1} \end{bmatrix} = \begin{bmatrix} \hat{x}_t \\ \hat{u}_{t-1} \end{bmatrix} + [C_\xi \quad C_w] \begin{bmatrix} \xi_t - \hat{\xi}_t \\ \omega_t \end{bmatrix} \in \begin{bmatrix} \hat{x}_t \\ \hat{u}_{t-1} \end{bmatrix} \oplus [C_\xi \quad C_w] \tilde{Z}. \quad (15)$$

Therefore, in order to guarantee that $(x_t, u_{t-1}) \in X \times U$, the following tightened constraints must be satisfied for $(\hat{x}_t, \hat{u}_{t-1})$:

$$(\hat{x}_t, \hat{u}_{t-1}) \in \hat{X}_U \quad (16)$$

where $\hat{X}_U$ is the set, with non-empty interior, that verifies

$$\hat{X}_U \subseteq (X \times U) \ominus [C_\xi \quad C_w] \tilde{Z}. \quad (17)$$

In terms of $\xi, \hat{r}$ and in view of (11), (16) can be written as

$$C^* \begin{bmatrix} \hat{\xi}_t \\ \hat{r}_t \end{bmatrix} \in \hat{X}_U. \quad (18)$$

C. The Maximal Output Admissible Set

In order to design an MPC stabilizing feedback control law, it is advisable to define the auxiliary control law

$$\delta \hat{u}_t = \mathcal{K} \hat{\xi}_t \quad (19)$$

to compute an invariant set for the nominal closed-loop system (7), (19), together with a set of output reference values $\hat{r}$, where $(\hat{\xi}_t, \hat{r})$ must lie in order to guarantee that constraints $(\hat{x}_t, \hat{u}_{t-1}) \in \hat{X}_U$ are verified for all t . This requires to compute the maximal output admissible set (MOAS) $\mathcal{O}$ for the auxiliary system with the dynamic equation

$$\begin{bmatrix} \hat{\xi}_{t+1} \\ \hat{r} \end{bmatrix} = \begin{bmatrix} \mathcal{F} & 0 \\ 0 & I_m \end{bmatrix} \begin{bmatrix} \hat{\xi}_t \\ \hat{r} \end{bmatrix} \quad (20)$$

and the output (11). To this end, consider the following

Assumption 2:

- i) the pair $(\begin{bmatrix} \mathcal{F} & 0 \\ 0 & I_m \end{bmatrix}, C^*)$ is observable,
- ii) $\hat{X}_U$ is a closed polytope,

Under this assumption, an invariant, polytopic inner approximation $\mathcal{O}_\varepsilon$ to the MOAS can be computed in a finite number of steps as shown in [6]. Specifically, $\mathcal{O}_\varepsilon$ is defined as follows:

$$\mathcal{O}_\varepsilon = \left\{ (\hat{\xi}, \hat{r}) \in \mathbb{R}^{n+2m} : C_\xi \mathcal{F}^t \hat{\xi} + C_y \hat{r} \in \hat{X}_U \text{ for all } t \geq 0 \text{ and } C_y \hat{r} \in \hat{X}_U(\varepsilon) \right\} \quad (21)$$

where $\hat{X}_U(\varepsilon)$ is a close and compact set satisfying $\hat{X}_U(\varepsilon) \oplus \mathcal{B}_\varepsilon^{n+m}(0) \subseteq \hat{X}_U$, and ε can be arbitrarily small. Note that $\mathcal{O}_\varepsilon \subset \mathcal{O}$, see again [6].

It is finally worth remarking that, in view of (21), a reference output $\hat{r}$ is feasible if $C_y \hat{r} \in \hat{X}_U(\varepsilon)$.

Authorized licensed use limited to: SUN YAT-SEN UNIVERSITY. Downloaded on March 20, 2024 at 10:22:01 UTC from IEEE Xplore. Restrictions apply.

III. THE MPC PROBLEM

A. The Case of Bounded Variable Disturbance

In the formulation of the robust MPC problem and according to [14], at any time t the optimization problem is stated with reference to the nominal system (7) and its initial condition $\hat{\xi}_t$ is considered as an optimization variable, besides the future nominal input sequence $\delta \hat{u}_{[t:t+N-1]}$. Being $\hat{\xi}_t = (\delta \hat{x}_t, (\hat{y}_t - \hat{r}_t))$, this means that both $\delta \hat{x}_t$ and $\hat{y}_t$ are arguments of the optimization problem. As for possibly infeasible reference signals r^o , and following [9], the value $\hat{r}_t$ to be considered as the output reference trajectory in the MPC problem at time t is computed as the nearest point to r^o , which at the same time guarantees feasibility. As such, also $\hat{r}_t$ is regarded as an argument of the optimization problem itself, rather than a fixed parameter.

In conclusion, the MPC problem to be solved at each time instant t is

$$\begin{aligned} V_N^*(r^o, \delta x_t, y_t) \\ = \min_{\delta \hat{x}_t, \hat{y}_t, \hat{r}_t, \delta \hat{u}_{[t:t+N-1]}} V_N(\delta \hat{x}_t, \hat{y}_t, \hat{r}_t, \delta \hat{u}_{[t:t+N-1]}; r^o, \delta x_t, y_t) \end{aligned} \quad (22)$$

where

$$V_N = \|\hat{r}_t - r^o\|_T^2 + \sum_{k=t}^{t+N-1} \left\{ \|\hat{\xi}_k\|_Q^2 + \|\delta \hat{u}_k\|_R^2 \right\} + \|\hat{\xi}_{t+N}\|_P^2$$

subject to the dynamic constraint (7), to the constraints

$$\begin{bmatrix} I_{n+m} \\ 0 \end{bmatrix} (\xi_t - \hat{\xi}_t) \in \begin{bmatrix} 0 \\ -w_{t-1} \end{bmatrix} \oplus \tilde{Z} \quad (23)$$

where, at time t , w_{t-1} can be explicitly computed from x_{t-1} , u_{t-1} , and x_t using (1)

$$C^* \begin{bmatrix} \hat{\xi}_{t+k} \\ \hat{r}_t \end{bmatrix} \in \hat{X}_U \quad (24)$$

for all $k = 1, \dots, N-1$, and to the terminal constraint

$$\begin{bmatrix} \hat{\xi}_{t+N} \\ \hat{r}_t \end{bmatrix} \in \mathcal{O}_\varepsilon. \quad (25)$$

According to the receding horizon approach, at any time instant the optimal solution $\delta \hat{x}_{t|t}, \hat{y}_{t|t}, \hat{r}_{t|t}, \delta \hat{u}_{[t:t+N-1]|t}$ is found and the control law (8) is applied with $\delta \hat{u}_t = \delta \hat{u}_{t|t}$ and $\hat{\xi}_t = (\delta \hat{x}_{t|t}, (\hat{y}_{t|t} - \hat{r}_{t|t}))$.

A proper selection of the tuning parameters allows one to prove the convergence properties of the proposed control algorithm. In particular, the weighting matrices Q and R are assumed to be symmetric and positive definite, while P is to be chosen as the positive definite solution of the Lyapunov equation

$$\mathcal{F}^T P \mathcal{F} - P = -(Q + \mathcal{K}^T R \mathcal{K}). \quad (26)$$

As for the matrix T , decompose $P \in \mathbb{R}^{(n+m) \times (n+m)}$ as follows:

$$P = \begin{bmatrix} P_{xx} & P_{xy} \\ P_{xy}^T & P_{yy} \end{bmatrix}$$

where $P_{xx} \in \mathbb{R}^{n \times n}$, and select T to satisfy the inequality

$$T - P_{yy} \succ 0. \quad (27)$$

The selection of T guaranteeing optimal closed-loop performances is addressed to in [5]. With these choices, the main convergence result can now be proved along the lines depicted in [9] and [2].

Theorem 1: Let Assumption 1 be verified and the design parameters $Q, R, P, T, \tilde{Z}, \hat{\mathbb{X}}_{\mathbb{U}}$, and $\mathbb{O}_{\varepsilon}$ be chosen as specified. Then, if at time $t = 0$ a feasible solution to the optimization problem (22)–(25) exists, the resulting MPC control law asymptotically steers the nominal system output $\hat{y}_t$ to the admissible set-point r_{ad} , where

$$r_{ad} = \arg \min_{C_y y \in \hat{\mathbb{X}}_{\mathbb{U}}(\varepsilon)} \|y - r^o\|_T^2. \quad (28)$$

Moreover, $\delta \hat{u}_t \rightarrow 0$ as $t \rightarrow \infty$, and the constraints $(x_t, u_t) \in \mathbb{X} \times \mathbb{U}$ are fulfilled for all $t \geq 0$. ■

This result guarantees that, for any feasible initial state, the proposed method steers the nominal system output to the desired reference value r^o if r^o is admissible, or to the admissible output r_{ad} if r^o is not admissible. Since the controller always keeps $\xi_k - \hat{\xi}_k$ bounded for all $k \geq 0$, in view of (13), the output of the real system is driven in a neighborhood of r^o (or of r_{ad}). Moreover, the following corollary accounts for the case the disturbance converges to a constant value.

Corollary 1: Let Assumption 1 be verified, the parameters $Q, R, P, T, \tilde{Z}, \hat{\mathbb{X}}_{\mathbb{U}}$, and $\mathbb{O}_{\varepsilon}$ be chosen as specified. If $w_t \rightarrow \bar{w}$ as $t \rightarrow \infty$ and if, at time $t = 0$, a feasible solution to the optimization problem (22)–(25) exists, then the resulting MPC control law asymptotically steers the output y_t of system (1) to the admissible set-point r_{ad} given by (28) and the constraints $(x_t, u_t) \in \mathbb{X} \times \mathbb{U}$ are fulfilled for all $t \geq 0$. ■

B. The Case of Constant Unknown Disturbance

When the disturbance $w_t \in \mathbb{W}$ is unknown but constant, i.e., $w_t = \bar{w}$, a standard and much simpler formulation of the MPC problem can be used. In fact, being $\delta w_t = 0$, the velocity forms of the system (4) and of its nominal model (7) coincide. Therefore $\tilde{Z} = \{0\} \times \mathbb{W}$ which means that, provided that $\xi_t = \hat{\xi}_t$, $\xi_{t+j} = \hat{\xi}_{t+j}$ for all $\bar{w} \in \mathbb{W}$. In view of this, the robust “tube-based” MPC formulation is no longer needed and the optimization problem can be directly formulated on the plant variables ξ_t and δu_t . Moreover, to guarantee $(x_t, u_{t-1}) \in \mathbb{X} \times \mathbb{U}$, the set $\hat{\mathbb{X}}_{\mathbb{U}}$ in (16), (18), and (21) must verify

$$\hat{\mathbb{X}}_{\mathbb{U}} \subseteq (\mathbb{X} \times \mathbb{U}) \oplus C_w \mathbb{W} \quad (29)$$

which is the reformulation of (17) in this simpler case. In summary, the optimization problem to be solved online has the following standard formulation of (non-robust) MPC:

$$V_N^*(r^o, \delta x_t, y_t) = \min_{\hat{r}_t, \delta u_{[t:t+N-1]}} V_N(\hat{r}_t, \delta u_{[t:t+N-1]}; r^o, \delta x_t, y_t) \quad (30)$$

where

$$V_N = \|\hat{r}_t - r^o\|_T^2 + \sum_{k=t}^{t+N-1} \left\{ \|\xi_k\|_Q^2 + \|\delta u_k\|_R^2 \right\} + \|\xi_{t+N}\|_P^2$$

subject to the dynamic constraint (4) with $d_t = 0$, to the constraints (24), for all $k = 1, \dots, N-1$, and to the terminal constraint (25).

Once the optimization problem has been solved at any time t and its optimal solution $\hat{r}_{t|t}, \delta u_{[t:t+N-1]|t}$ has been computed, the input applied to the real system is $u_t = u_{t-1} + \delta u_{t|t}$.

The following corollary of Theorem 1 can be stated.

Corollary 2: Let Assumption 1 be verified, the design parameters $Q, R, P, T, \tilde{Z}, \hat{\mathbb{X}}_{\mathbb{U}}$, and $\mathbb{O}_{\varepsilon}$ be chosen as specified. Consider the case of constant disturbances, i.e., $w_t = \bar{w}$ for all $t \geq 0$. Then, if at time $t = 0$ a feasible solution to the optimization problem (30) exists, the resulting MPC control law asymptotically steers the output y_t of system (1) to the admissible set-point r_{ad} given by (28) and the constraints $(x_t, u_t) \in \mathbb{X} \times \mathbb{U}$ are fulfilled for all $t \geq 0$. ■

It is worth recalling that, in the definition (28) of r_{ad} , the set $\hat{\mathbb{X}}_{\mathbb{U}}(\varepsilon)$ appears, which is defined on the basis $\hat{\mathbb{X}}_{\mathbb{U}}$, which now must verify the condition (29).

C. The Case of Model Uncertainties

A possible application of the results presented in this technical note is the case where the disturbance w_t stems from model perturbations. Specifically, assume that the real system matrices are $A_r = A + A_{\delta}$ and $B_r = B + B_{\delta}$, with A_{δ} and B_{δ} unknown, but bounded. Since $x_t \in \mathbb{X}$ and $u_t \in \mathbb{U}$, the control problem can be stated as in (1), where the bounded disturbance is

$$w_t = A_{\delta} x_t + B_{\delta} u_t. \quad (31)$$

We assume, according to [7], that

$$[A_{\delta} \ B_{\delta}] = \sum_{i=1}^{n_{\alpha}} \alpha_i [A_i \ B_i] \quad (32)$$

where $\alpha_i > 0$ are unknown time-invariant parameters satisfying $\sum_{i=1}^{n_{\alpha}} \alpha_i = 1$. Define also

$$A_{\delta} = \begin{bmatrix} A_{\delta} & 0 \\ C A_{\delta} & 0 \end{bmatrix} \quad B_{\delta} = \begin{bmatrix} B_{\delta} \\ C B_{\delta} \end{bmatrix}.$$

The following corollary of Theorem 1 can be stated, provided that all the ingredients of the MPC problem (22) can be properly defined, including $\hat{\mathbb{X}}_{\mathbb{U}}$.

Corollary 3: Let Assumption 1 be verified, the design parameters $Q, R, P, T, \tilde{Z}, \hat{\mathbb{X}}_{\mathbb{U}}$, and $\mathbb{O}_{\varepsilon}$ be chosen as specified, and model uncertainties be of the type (31). Furthermore, assume that the matrix $\mathcal{F}_r = A + A_{\delta} + (B + B_{\delta})\mathcal{K}$ is Schur for all pairs $[A_{\delta} \ B_{\delta}]$ satisfying (32). Then, if at time $t = 0$ a feasible solution to the optimization problem (22)–(25) exists, the resulting MPC control law asymptotically steers the output y_t of system (1) to the admissible set-point r_{ad} given by (28) and the constraints $(x_t, u_t) \in \mathbb{X} \times \mathbb{U}$ are fulfilled for all $t \geq 0$. ■

Methods for designing the gain matrix $\mathcal{K}$ satisfying the assumptions of the corollary have been thoroughly studied in the past, also in the context of MPC, see, e.g., [4], [7].

One of the key points of the proposed approach lies in the existence of a proper set $\hat{\mathbb{X}}_{\mathbb{U}}$. According to its definition, it is computed as follows: (I) compute $W = \text{convH}(\bigcup_{i=1}^{n_{\alpha}} \{[A_i \ B_i] \mathbb{X} \times \mathbb{U}\})$, where convH denotes the *convex hull*; (II) compute $\tilde{Z}$ using (14); (III) compute $\hat{\mathbb{X}}_{\mathbb{U}}$ using (17). Note that step (III) can be carried out provided that

$$[C_{\xi} \ C_w] \tilde{Z} \subset \mathbb{X} \times \mathbb{U} \quad (33)$$

which, in view of the fact that $\tilde{Z}$ is in turn computed from $\mathbb{X} \times \mathbb{U}$, is indeed a *small gain* requirement. In fact, small perturbations (i.e., small A_{δ} and B_{δ}) result in a small $\tilde{Z}$. A simple sufficient condition guaranteeing (33) can be stated in case $\mathbb{X} \times \mathbb{U}$ is a zonotope i.e., a centrally symmetric convex polytope, defined in the following alternative ways:

$$\begin{aligned} \mathbb{X} \times \mathbb{U} &= \left\{ (x, u) \in \mathbb{R}^{n+m} : \begin{bmatrix} x \\ u \end{bmatrix} = \Xi_{x,u} v \text{ where } \|v\|_{\infty} \leq 1 \right\} \\ &= \left\{ (x, u) \in \mathbb{R}^{n+m} : h_i^T \begin{bmatrix} x \\ u \end{bmatrix} \leq 1 \text{ for all } i = 1, \dots, r \right\} \end{aligned} \quad (34)$$

for suitably-specified matrices $\Xi_{x,u}$ and vectors $h_i, i = 1, \dots, r$. Also define $H_{x,u}$ as

$$H_{x,u} = [h_1 \ \dots \ h_r]. \quad (35)$$

The following proposition provides a sufficient (but conservative) condition for the small gain one (33).

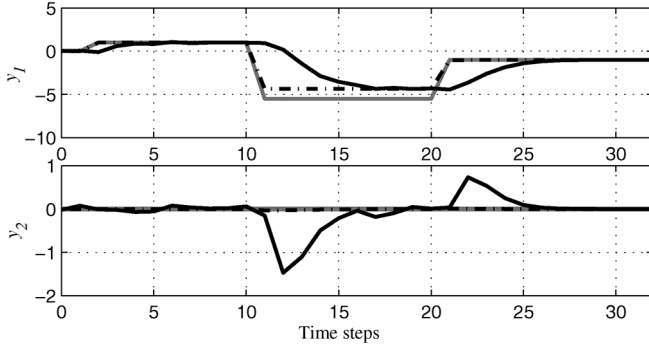


Fig. 1. Trajectories of the real plant output variables y_1 (above) and y_2 (below). Grey solid line: real desired references ($r_{1,2}^o$); black dash-dotted line: artificial references ($\hat{r}_{1,2}$); black solid line: outputs of the system ($y_{1,2}$). The disturbance w_t is constant for $t \geq 21$.

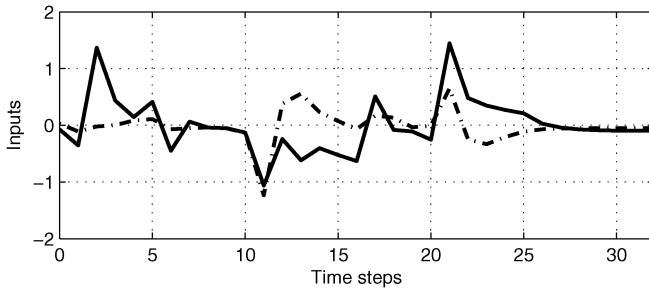


Fig. 2. Trajectories of the inputs applied to the real system. Black solid line: first input; black dash-dotted line: second input.

Proposition 3: Assume that

$$\sum_{k=0}^{\infty} \|\tilde{\mathcal{F}}^k\|_{\infty} \|H_{x,u}^T [C_{\xi} \ C_w]\|_{\infty} \max_{j=1,\dots,n_{\alpha}} \|\tilde{B}_w [A_j \ B_j] \Xi_{x,u}\|_{\infty} < 1 \quad (36)$$

then $\hat{\mathbb{X}}_{\cup}$ can be properly defined. ■

Checking (36) is straightforward: $H_{x,u}$ and $\Xi_{x,u}$ are given once sets $\mathbb{X}$ and $\cup$ are defined, and all the matrices appearing in (36) are defined in the problem setup. Furthermore, since $\tilde{\mathcal{F}}$ is Schur, there exist constants $\alpha \geq 1$, $\beta \in [0, 1)$ such that $\|\tilde{\mathcal{F}}^k\|_{\infty} \leq \alpha\beta^k$. Therefore, by replacing $\sum_{k=0}^{\infty} \|\tilde{\mathcal{F}}^k\|_{\infty}$ with $\alpha/(1-\beta)$, we obtain a sufficient condition for (36).

IV. EXAMPLE

Consider the following benchmark problem, inspired by [8]–[10]. The model (1) is characterized by $C = I$

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0.47 - 0.06(\alpha_1 - \alpha_2) & 0 \\ 0 & 1.03 + 0.06(\alpha_1 - \alpha_2) \end{bmatrix}$$

where uncertainties affect B and $\alpha_1 = 1 - \alpha_2 = 0.25$. The constraints on states and inputs are $\|x\|_{\infty} \leq 5$ and $\|u\|_{\infty} \leq 3$, while the disturbance w is such that $\|w\|_{\infty} \leq 0.1$. For the synthesis of the controller it has been set $N = 5$, $Q = I_4$, $R = 0.01I_2$, while the matrices K , P have been computed with the LQ synthesis criterion. The weighting matrix T has been set to $T = 10P_{yy}$.

In the reported simulations, the reference trajectory for y_2 is constant and equal to zero. The first output y_1 , instead, has a piecewise constant reference trajectory, taking values 5, -5.5 and 3. The disturbance is randomly varying for $t \in [0, 20]$, while it is kept constant for $t \in [21, 32]$. The results are shown in Fig. 1, while Fig. 2 shows the

trajectories of the input variables. Note that every admissible reference signal is tracked, without error when $w_t = \bar{w} = 0.05$, and that the effects of unknown disturbances and model uncertainties are rejected. On the other hand, the setpoint -5.5 is not admissible and the output is steered to the closest admissible value.

V. CONCLUSION

In this technical note, a robust MPC algorithm solving the offset-free tracking and the infeasible reference problems has been developed for disturbed systems expressed in velocity form. Convergence results have been obtained by suitably defining the auxiliary control law and the terminal set used in the problem formulation. Further research will be devoted to extend this approach to the case of distributed systems.

APPENDIX A

PROOF OF PROPOSITION 2

Being $\delta x_t = x_t - x_{t-1}$, system (1) can be written as

$$\delta x_t = (A - I_n)x_{t-1} + Bu_{t-1} + w_{t-1} \quad (37)$$

$$y_t = Cx_t = C(Ax_{t-1} + Bu_{t-1} + w_{t-1})$$

which gives

$$\begin{bmatrix} x_{t-1} \\ u_{t-1} \end{bmatrix} = \begin{bmatrix} A - I_n & B \\ CA & CB \end{bmatrix}^{-1} \left(\begin{bmatrix} \delta x_t \\ y_t \end{bmatrix} - \begin{bmatrix} I_n \\ C \end{bmatrix} w_{t-1} \right). \quad (38)$$

The dynamics of the state of system (1) allows one to write also

$$\begin{bmatrix} x_t \\ u_{t-1} \end{bmatrix} = \begin{bmatrix} A & B \\ 0 & I_m \end{bmatrix} \begin{bmatrix} x_{t-1} \\ u_{t-1} \end{bmatrix} + \begin{bmatrix} I_n \\ 0 \end{bmatrix} w_{t-1}. \quad (39)$$

Inserting (38) inside (39) and recalling that

$$\begin{bmatrix} \delta x_t \\ y_t \end{bmatrix} = \begin{bmatrix} I_n & 0 & 0 \\ 0 & I_m & I_m \end{bmatrix} \begin{bmatrix} \xi_t \\ \hat{r}_t \end{bmatrix}$$

(10) clearly follows. Expression (11) is obtained in the same way simply considering $w_{t-1} = 0$.

In order to find matrices C^* and C_w we need to prove that the inverse of Σ can be computed. It is easy to see that Σ is invertible if and only if S is invertible, which holds in view of Assumption 1.

APPENDIX B

PROOF OF THEOREM 1

The proof of Theorem 1 follows the same rationale as the proof of Theorem 1 in [9]. Specifically, it is based on the following steps:

- I) we prove that the only closed-loop stable equilibrium point such that $\hat{\xi}_t = 0$ and compatible with (22) is the one corresponding to $\hat{r}_t = r_{ad}$;
- II) we prove recursive feasibility and that the system converges asymptotically to an equilibrium point, i.e., that $\hat{\xi}_t \rightarrow 0$ as $t \rightarrow \infty$;
- III) from the previous steps we infer that the nominal system asymptotically converges to the unique equilibrium point which is compatible with (22).

Step I: First note that, among all admissible points $(\hat{\xi}_t, \hat{r}_t) = (\delta \hat{x}_t, \hat{y}_t - \hat{r}_t, \hat{r}_t)$, the point $(0, 0, r_{ad})$, where r_{ad} fulfills (28), is the equilibrium condition to (7) (corresponding to $\hat{y}_t = \hat{r}_t = r_{ad}$ and $\delta \hat{u}_k = 0$ for all k) such that the cost function V_N is globally minimized. Recall that it is admissible since $C_y r_{ad} \in \hat{\mathbb{X}}_{\cup}(\epsilon)$, see (21).

Consider a generic admissible reference $\hat{r} \neq r_{ad}$, such that $(\hat{x}_r, \hat{u}_r) = C_y \hat{r} \in \hat{\mathbb{X}}_{\cup}(\epsilon)$, corresponding to the equilibrium $(\delta \hat{x}_t, \hat{y}_t - \hat{r}_t, \hat{r}_t) = (0, 0, \hat{r})$ for the system (20) (i.e., corresponding

to an equilibrium to (7) in case $\delta \hat{u}_k = 0$ for all $k \geq t$). The cost, associated to such equilibrium condition is

$$V_N = \|\hat{r} - r^o\|_T^2 = \|\hat{r} - r_{ad}\|_T^2 + \|r_{ad} - r^o\|_T^2 + 2(r_{ad} - r^o)^T T(\hat{r} - r_{ad}). \quad (40)$$

Instead, consider the solution corresponding to the trajectory starting from $(\delta \hat{x}_t, \hat{y}_t - \hat{r}_t, \hat{r}_t) = (0, \hat{r} - \tilde{r}, \tilde{r})$ (i.e., from the point $(\hat{x}_r, \hat{u}_r)$, which does not correspond now to an equilibrium point for (6), in view of the fact that the reference output is now $\tilde{r} \neq \hat{r}$), and evolving according to the auxiliary control law (19). If $\tilde{r}$ is defined according to $\tilde{r} = \lambda \hat{r} + (1 - \lambda)r_{ad}$, $\lambda \in [0, 1]$, then

- a) the corresponding equilibrium is admissible, i.e., $C_y \tilde{r} \in \hat{\mathbb{X}}_U(\epsilon)$, in view of the convexity of $\hat{\mathbb{X}}_U(\epsilon)$
- b) since $C_y \tilde{r} \in \hat{\mathbb{X}}_U(\epsilon)$, and being $\mathbb{O}_\epsilon \subset \mathbb{O}$, provided $(1 - \lambda)$ (and hence $\|\hat{r} - \tilde{r}\|$) is sufficiently small, we get that the initial condition $(0, \hat{r} - \tilde{r}, \tilde{r}) \in \mathbb{O}$. In this way, it is possible to control the system with the auxiliary law (19).

The cost associated to such auxiliary control law is

$$\tilde{V}_N = \|\hat{r} - r^o\|_T^2 + \sum_{k=0}^{N-1} \left\{ \left\| \mathcal{F}^k(0, \hat{r} - \tilde{r}) \right\|_Q^2 + \left\| \mathcal{K} \mathcal{F}^k(0, \hat{r} - \tilde{r}) \right\|_R^2 \right\} + \left\| \mathcal{F}^N(0, \hat{r} - \tilde{r}) \right\|_P^2. \quad (41)$$

In view of (26), we obtain that the latter is equal to

$$\tilde{V}_N = \|\hat{r} - r^o\|_T^2 + \|(0, \hat{r} - \tilde{r})\|_P^2 = \|\hat{r} - r^o\|_T^2 + \|\hat{r} - \tilde{r}\|_{P_{yy}}^2 \quad (42)$$

where

$$\|\hat{r} - r^o\|_T^2 = \|\hat{r} - r_{ad}\|_T^2 + \|r_{ad} - r^o\|_T^2 + 2(r_{ad} - r^o)^T T(\hat{r} - r_{ad}).$$

Since $P_{yy} \prec T$

$$V_N - \tilde{V}_N \geq \|\hat{r} - r_{ad}\|_T^2 - \|\hat{r} - r_{ad}\|_T^2 - \|\hat{r} - \tilde{r}\|_{P_{yy}}^2 + 2(r_{ad} - r^o)^T T(\hat{r} - \tilde{r}). \quad (43)$$

Note that $a = \|\hat{r} - r_{ad}\|_T = \lambda \|\hat{r} - r_{ad}\|_T$, $b = \|\hat{r} - \tilde{r}\|_T = (1 - \lambda) \|\hat{r} - r_{ad}\|_T$, and $c = a + b = \|\hat{r} - r_{ad}\|_T$. Since $a, b, c > 0$, if $a + b = c$ then

$$a^2 + b^2 < c^2. \quad (44)$$

Defining $f(r) = \|r - r^o\|_T^2$, we have that

$$2(r_{ad} - r^o)^T T(\hat{r} - \tilde{r}) = (1 - \lambda) \left. \frac{df}{dr} \right|_{r=r_{ad}} (\hat{r} - r_{ad}) \geq 0 \quad (45)$$

in view of the optimality of r_{ad} with respect to $f(r)$ in the admissible set. From (43), (44), and (45)

$$\tilde{V}_N < V_N$$

from which it follows that, for all $\hat{r} \neq r_{ad}$, $(0, 0, \hat{r})$ is not an equilibrium point for the closed loop system.

Step II: Recursive feasibility and convergence of $\hat{\xi}_t$ to zero, once $\hat{r}$ is kept constant, can be proved resorting to standard robust tube-based MPC arguments, see [14].

Feasibility: Consider that, at time t , a solution to (22) is $(\delta \hat{x}_t, \hat{y}_t, \hat{r}_t, \delta \hat{u}_{[t:t+N-1]|t})$. Then it is easy to see that a feasible solution to (22) at time $t + 1$ is $(\delta \hat{x}_{t+1}, \hat{y}_{t+1}, \hat{r}_t, \delta \hat{u}_{[t+1:t+N]|t})$, where it is defined $\delta \hat{u}_{t+N|t} = \mathcal{K} \hat{\xi}_{t+N|t}$, where $\hat{\xi}_{t+N|t}$ stems

from (7) with inputs $\delta \hat{u}_{[t:t+N-1]|t}$ and initial condition $\hat{\xi}_{t|t} = (\delta \hat{x}_{t|t}, (\hat{y}_{t|t} - \hat{r}_{t|t}))$.

Convergence: Consider the above-mentioned feasible solution at time $t + 1$. Following standard arguments for the proof of the convergence of robust tube-based MPC [14] we obtain that

$$V_N^*(r^o, \delta x_{t+1}, y_{t+1}) - V_N^*(r^o, \delta x_t, y_t) \leq -\|\hat{\xi}_{t|t}\|_Q^2 - \|\delta \hat{u}_{t|t}\|_R^2.$$

In view of the positive definiteness of V_N^* and of Q and R , it holds that $\hat{\xi}_{t|t} \rightarrow 0$ and $\delta \hat{u}_{t|t} \rightarrow 0$ as $t \rightarrow \infty$.

Step III: In view of Step II, convergence of $\hat{\xi}_t$ is guaranteed. Since, in view of Step I, the only equilibrium to (7) compatible with (22) is the one corresponding to $\hat{r}_t = r_{ad}$, then it also holds that $\hat{y}_t \rightarrow r_{ad}$ as $t \rightarrow \infty$.

APPENDIX C PROOF OF THE COROLLARIES

Proof of Corollary 1: In view of Theorem 1 $\hat{y}_t \rightarrow r_{ad}$ as $t \rightarrow \infty$. Recalling (4), since the input increment to the real system is $\delta u_t = \delta \hat{u}_{t|t} + \mathcal{K}(\xi_t - \hat{\xi}_{t|t})$, the system dynamics is described by $\xi_{t+1} = \mathcal{F} \xi_t + \delta \hat{u}_{t|t} - \mathcal{K} \hat{\xi}_{t|t} + d_t$ where both $\delta \hat{u}_{t|t} - \mathcal{K} \hat{\xi}_{t|t}$ and $d_t = \mathcal{B}_w(w_t - w_{t-1})$ are asymptotically vanishing. Being $\mathcal{F}$ Schur stable then $\xi_t \rightarrow 0$ as $t \rightarrow \infty$. ■

Proof of Corollary 2: In view of Theorem 1 $\hat{y}_t \rightarrow r_{ad}$ as $t \rightarrow \infty$. Recalling that, in this particular case, $\tilde{\mathbb{Z}} = \{0\} \times \mathbb{W}$, we have that

$$y_t \rightarrow r_{ad} \oplus [0 \quad I_m \quad 0] \tilde{\mathbb{Z}} = r_{ad} \oplus \{0\}$$

as $t \rightarrow \infty$. ■

Proof of Corollary 3: In view of Theorem 1, it is proved that $\hat{y}_t \rightarrow r_{ad}$ as $t \rightarrow \infty$. We define $\varepsilon_t^a = y_t - r_{ad}$ and $\varepsilon_t^b = \hat{y}_t - r_{ad}$. Noting that $\delta w_t = w_t - w_{t-1} = A_\delta \delta x_t + B_\delta \delta u_t$ we have that, similarly to (2)

$$\begin{aligned} \delta x_{t+1} &= (A + A_\delta) \delta x_t + (B + B_\delta) \delta u_t \\ \varepsilon_{t+1}^a &= C(A + A_\delta) \delta x_t + \varepsilon_t^a + C(B + B_\delta) \delta u_t \end{aligned} \quad (46)$$

where, for all t , $\delta u_t = \delta \hat{u}_{t|t} + \mathcal{K} \left[\begin{smallmatrix} \delta x_t - \delta \hat{x}_{t|t} \\ \varepsilon_t^a - \varepsilon_t^b \end{smallmatrix} \right]$. We can therefore write (46) as

$$\begin{bmatrix} \delta x_{t+1} \\ \varepsilon_{t+1}^a \end{bmatrix} = \mathcal{F}_r \begin{bmatrix} \delta x_t \\ \varepsilon_t^a \end{bmatrix} + \eta_t \quad (47)$$

where $\eta_t = \begin{bmatrix} B+B_\delta \\ C(B+B_\delta) \end{bmatrix} (\delta \hat{u}_{t|t} - \mathcal{K} \left[\begin{smallmatrix} \delta x_t - \delta \hat{x}_{t|t} \\ \varepsilon_t^a - \varepsilon_t^b \end{smallmatrix} \right])$ is an asymptotically vanishing term. Since $\mathcal{F}_r$ is Schur, it implies that $\delta x_t \rightarrow 0$ and $\varepsilon_t^a \rightarrow 0$ as $t \rightarrow \infty$.

APPENDIX D PROOF OF PROPOSITION 3

Recalling (34) and (17) $\hat{\mathbb{X}}_U$ exists if there exists $\rho > 0$ such that

$$[C_\xi \quad C_w] \tilde{\mathbb{Z}} \oplus \mathcal{B}_\rho^{(2n+m)}(0) \subseteq \mathbb{X} \times \mathbb{U} \quad (48)$$

which is verified if, for all $i = 1, \dots, r$

$$\max_{\tilde{z} \in \tilde{\mathbb{Z}}} h_i^T [C_\xi \quad C_w] \tilde{z} < 1. \quad (49)$$

In view of (14), (49) is verified if, for all $i = 1, \dots, r$

$$\sum_{k=0}^{\infty} \max_{w \in \mathbb{W}} h_i^T [C_\xi \quad C_w] \tilde{\mathcal{F}}^k \tilde{\mathcal{B}}_w w < 1 \quad (50)$$

where, in view of (34), $w = \sum_{j=1}^{n_\alpha} \alpha_j [A_j \ B_j] \Xi_{x,u} v$, where $\|v\|_\infty \leq 1$. Therefore (50) is verified if, for all $i = 1, \dots, r$ and for all α_j , $j = 1, \dots, n_{alpha}$

$$\sum_{k=0}^{\infty} \max_{\|v\|_\infty \leq 1} h_i^T [C_\xi \ C_w] \tilde{\mathcal{F}}^k \tilde{\mathcal{B}}_w \sum_{j=1}^{n_\alpha} \alpha_j [A_j \ B_j] \Xi_{x,u} v < 1. \quad (51)$$

Note that, for all $i = 1, \dots, r$

$$\begin{aligned} \sum_{k=0}^{\infty} \max_{\|v\|_\infty \leq 1} h_i^T [C_\xi \ C_w] \tilde{\mathcal{F}}^k \tilde{\mathcal{B}}_w \sum_{j=1}^{n_\alpha} \alpha_j [A_j \ B_j] \Xi_{x,u} v \\ = \sum_{k=0}^{\infty} \left\| h_i^T [C_\xi \ C_w] \tilde{\mathcal{F}}^k \tilde{\mathcal{B}}_w \sum_{j=1}^{n_\alpha} \alpha_j [A_j \ B_j] \Xi_{x,u} \right\|_\infty \end{aligned}$$

Since, in view of (35) and for all $k = 0, \dots, \infty$

$$\begin{aligned} \left\| h_i^T [C_\xi \ C_w] \tilde{\mathcal{F}}^k \tilde{\mathcal{B}}_w \sum_{j=1}^{n_\alpha} \alpha_j [A_j \ B_j] \Xi_{x,u} \right\|_\infty \\ \leq \left\| H_{x,u}^T [C_\xi \ C_w] \tilde{\mathcal{F}}^k \tilde{\mathcal{B}}_w \sum_{j=1}^{n_\alpha} \alpha_j [A_j \ B_j] \Xi_{x,u} \right\|_\infty \end{aligned}$$

for all $i = 1, \dots, r$, then

$$\begin{aligned} \sum_{k=0}^{\infty} \left\| h_i^T [C_\xi \ C_w] \tilde{\mathcal{F}}^k \tilde{\mathcal{B}}_w \sum_{j=1}^{n_\alpha} \alpha_j [A_j \ B_j] \Xi_{x,u} \right\|_\infty \\ \leq \sum_{k=0}^{\infty} \left\| H_{x,u}^T [C_\xi \ C_w] \tilde{\mathcal{F}}^k \tilde{\mathcal{B}}_w \sum_{j=1}^{n_\alpha} \alpha_j [A_j \ B_j] \Xi_{x,u} \right\|_\infty \\ \leq \sum_{j=1}^{n_\alpha} \alpha_j \sum_{k=0}^{\infty} \left\| H_{x,u}^T [C_\xi \ C_w] \tilde{\mathcal{F}}^k \tilde{\mathcal{B}}_w [A_j \ B_j] \Xi_{x,u} \right\|_\infty \\ \leq \max_{j=1, \dots, n_\alpha} \left(\sum_{k=0}^{\infty} \left\| H_{x,u}^T [C_\xi \ C_w] \tilde{\mathcal{F}}^k \tilde{\mathcal{B}}_w [A_j \ B_j] \Xi_{x,u} \right\|_\infty \right) \\ \leq \sum_{k=0}^{\infty} \|\tilde{\mathcal{F}}^k\|_\infty \left\| H_{x,u}^T [C_\xi \ C_w] \right\|_\infty \cdot \max_{j=1, \dots, n_\alpha} \left\| \tilde{\mathcal{B}}_w [A_j \ B_j] \Xi_{x,u} \right\|_\infty. \end{aligned}$$

Eventually, if (36) is verified, it follows that (49) is also verified.

REFERENCES

- [1] I. Alvarado, D. Limon, T. Alamo, and E. F. Camacho, "Output feedback robust tube based MPC for tracking of piece-wise constant references," in *Proc. 46th IEEE Conf. Decision Control*, 2007, pp. 2175–2180.
- [2] G. Betti, M. Farina, and R. Scattolini, "An MPC algorithm for offset-free tracking of constant reference signals," in *Proc. 51st IEEE Conf. Decision Control*, 2012, pp. 5182–5187.
- [3] L. Chisci, A. Lombardi, and E. Mosca, "Dual receding horizon control of constrained discrete-time linear systems," *Eur. J. Control*, vol. 2, pp. 278–285, 1996.
- [4] F. A. Cuzzola, J. C. Geromel, and M. Morari, "An improved approach for constrained robust model predictive control," *Automatica*, vol. 38, pp. 1183–1189, 2002.
- [5] A. Ferramosca, D. Limon, I. Alvarado, T. Alamo, and E. F. Camacho, "MPC for tracking with optimal closed-loop performance," *Automatica*, vol. 45, no. 8, pp. 1975–1978, 2009.
- [6] E. G. Gilbert and K. T. Tan, "Linear systems with state and control constraints: The theory and application of maximal output admissible sets," *IEEE Trans. Autom. Control*, vol. 36, no. 9, pp. 1008–1020, Sep. 1991.
- [7] M. V. Kothare, V. Balakrishnan, and M. Morari, "Robust constrained model predictive control using linear matrix inequalities," *Automatica*, vol. 32, no. 10, pp. 1361–1379, 1996.
- [8] D. Limon, I. Alvarado, T. Alamo, and E. F. Camacho, "MPC for tracking of piece-wise constant references for constrained linear systems," in *Proc. 16th IFAC World Congress*, Prague, Czech Republic, 2005, pp. 135–140.
- [9] D. Limon, I. Alvarado, T. Alamo, and E. F. Camacho, "MPC for tracking piecewise constant references for constrained linear systems," *Automatica*, vol. 44, no. 9, pp. 2382–2387, 2008.
- [10] D. Limon, I. Alvarado, T. Alamo, and E. F. Camacho, "Robust tube-based MPC for tracking of constrained linear systems with additive disturbances," *J. Process Control*, vol. 20, no. 3, pp. 248–260, 2010.
- [11] U. Maeder, F. Borrelli, and M. Morari, "Nonlinear offset-free model predictive control," *Automatica*, vol. 48, no. 9, pp. 2059–2067, 2012.
- [12] L. Magni, G. De Nicolao, and R. Scattolini, "Output feedback and tracking of nonlinear systems with model predictive control," *Automatica*, vol. 37, pp. 1601–1607, 2001.
- [13] L. Magni and R. Scattolini, "Tracking of non-square nonlinear continuous time systems with piecewise constant model predictive control," *J. Process Control*, vol. 17, pp. 631–640, 2007.
- [14] D. Q. Mayne, M. M. Seron, and S. V. Raković, "Robust model predictive control of constrained linear systems with bounded disturbances," *Automatica*, vol. 41, no. 2, pp. 219–224, 2005.
- [15] M. Morari and U. Maeder, "Linear offset-free model predictive control," *Automatica*, vol. 45, no. 10, pp. 2214–2222, 2009.
- [16] K. R. Muske and T. A. Badgwell, "Disturbance modeling for offset-free linear model predictive control," *J. Process Control*, vol. 12, no. 5, pp. 617–632, 2002.
- [17] G. Pannocchia, "Robust disturbance modeling for model predictive control with application to multivariable ill-conditioned processes," *J. Process Control*, vol. 13, no. 8, pp. 693–701, 2003.
- [18] G. Pannocchia, "Robust model predictive control with guaranteed set-point tracking," *J. Process Control*, vol. 14, pp. 927–937, 2004.
- [19] G. Pannocchia and J. B. Rawlings, "Disturbance models for offset-free model predictive control," *AIChE J.*, vol. 49, no. 2, pp. 426–437, 2003.
- [20] G. Pannocchia and J. B. Rawlings, *The Velocity Algorithm LQR: A Survey Tech. Rep. 2001-01, 2001, TWMCC*.
- [21] D. M. Prett and C. E. Garcia, *Fundamental Process Control*. Stoneham, MA: Butterworths, 1988.
- [22] S. V. Raković, E. C. Kerrigan, K. I. Kouramas, and D. Q. Mayne, "Invariant approximations of the minimal robust positively invariant set," *IEEE Trans. Autom. Control*, vol. 50, no. 3, pp. 406–410, Mar. 2005.
- [23] J. B. Rawlings, "Tutorial overview of model predictive control," *IEEE Control Syst. Mag.*, vol. 20, no. 3, pp. 38–52, Mar. 2000.
- [24] L. Wang, "A tutorial on model predictive control: Using a linear velocity-form model," *Develop. Chem. Eng. Mineral Processing*, vol. 12, no. 5–6, pp. 573–614, 2004.